

SignTalk AI — Project Knowledge Base

Authoritative working reference for project understanding, development, presentation and viva

SignTalk AI | B.Tech CSE (AI/ML) | GLA University

1. Project Identity

Item	Details
Project	SignTalk AI: Real-Time Indian Sign Language Translation and Communication Platform
Technical subtitle	A Spatial-Temporal Graph and Transformer-Based Sign-to-Text System
Domain	AI / ML / Computer Vision / NLP
Primary implementation language	Indian Sign Language (ISL)
Primary output	Natural-language text captions
Product form	Web-based real-time application
Status	Academic project; clearly distinguish implemented, planned and future components

2. Problem

Sign-language users can face communication barriers with non-signers in healthcare, banking, public services, education, emergencies and customer support. Automated recognition is difficult because signing includes hand shapes, finger articulation, body posture, spatial relationships, facial/non-manual cues and temporal motion. Continuous cloud video can also introduce latency, bandwidth and privacy concerns.

3. Core Idea

Camera → OpenCV → MediaPipe Hands/Pose/Face → Landmark normalization and fusion → Graph construction → ST-GCN → Transformer/NLP → Real-time text.

4. Differentiators

- Continuous signing rather than isolated gesture classification.
- Hand/finger + body + facial/non-manual cues.
- Spatial-temporal graph modeling.
- Transformer-based sequence understanding and language generation.
- Target end-to-end latency below 500 ms.
- Privacy-aware, edge-capable architecture.
- Extensible multilingual architecture.

5. Scope

5.1 In Scope

- Webcam/video input
- OpenCV preprocessing

- MediaPipe hand landmarks
- Body pose/keypoints
- Facial features
- Feature normalization/fusion
- GCN/ST-GCN
- Transformer sequence modeling
- Sign-to-text
- NLP post-processing
- Real-time captions
- Offline inference design
- Web UI
- API/SDK architecture

5.2 Out of Scope

- Universal global sign-language support in v1
- Replacement of certified interpreters in legally mandated settings
- Medical/legal/financial advice
- Autonomous actions from translations
- Large-scale raw-video cloud storage
- Native mobile app in initial release

6. AI/ML Components

Component	Purpose
OpenCV	Capture, resize, normalize and manage video frames
MediaPipe Hands	Detailed hand/finger landmarks
Pose	Body movement and posture context
Face Mesh	Facial/non-manual cues
Feature fusion	Unified model-ready representation
GCN	Spatial relationships among joints
ST-GCN	Spatial relationships plus movement over time
Transformer	Sequence context and text generation
NLP	Sentence reconstruction and normalization

7. Why ST-GCN?

Landmarks form a natural graph: finger joints connect to the hand, the wrist connects to the arm, and body joints have structured relationships. GCN learns these spatial relationships; ST-GCN additionally learns how they change across frames. This is appropriate for movement-based signing.

8. Why Transformer?

Continuous signing is a sequence problem. A Transformer can model relationships across a sequence and support sequence-to-sequence generation. For example, a sequence representing 'I + hospital + tomorrow + go' may be reconstructed as 'I will go to the hospital tomorrow.' Exact output depends on the trained data and vocabulary.

9. Data Strategy

- Use an appropriate and legally usable ISL dataset and/or collected data.
- Extract frames/sequences and labels/transcriptions.
- Generate hand, body and facial landmarks.
- Normalize coordinates.
- Create temporal windows.
- Use justified augmentation.
- Prefer signer-independent evaluation splits where possible.
- Track dataset versions, label quality and class balance.

10. Training vs Real-Time Pipelines

Training: Dataset → preprocessing → landmarks → normalization → temporal sequences → ST-GCN → Transformer/NLP → evaluation.

Real time: Camera → preprocessing → landmarks → fusion → sliding window → ST-GCN → sequence representation → Transformer/NLP → live caption.

11. Metrics and Targets

Metric	Target/Measurement
End-to-end latency	<500 ms design target; experimentally benchmark
Recognition	Accuracy, precision, recall, F1
Translation	BLEU/ROUGE/semantic similarity where appropriate
FPS	Real-time runtime measurement
Robustness	Lighting, background, speed, occlusion, signer variation
Offline	Successful local inference on supported hardware
Usability	User testing
Privacy	Local-processing/network inspection

12. Experiments

- Hand-only vs hand+body vs hand+body+face.
- Baseline classifier/LSTM/GRU vs ST-GCN where feasible.
- Different signing speeds.
- Lighting and background changes.
- Camera distance/angle.
- Partial hand occlusion.
- Known vs unseen signers where data permits.

13. Security & Privacy

- Do not store raw video by default.
- Prefer local processing for sensitive environments.
- Keep secrets out of source control.
- Use authenticated protected APIs.

- Do not put raw video in logs.
- Define minimal retention for keypoints/session data.
- Respect dataset licenses and institutional policies.

14. Limitations

- Limited continuous ISL dataset diversity.
- Regional and signer variation.
- Hand occlusion and similar signs.
- Lighting/camera effects.
- Full unrestricted ISL is broader than a defined academic vocabulary.
- Sentence quality depends on training data.
- Edge performance depends on hardware and optimization.

15. Viva Essentials

Question	Answer
What is it?	A real-time AI system designed to translate supported continuous ISL into natural-language text.
What is the AI part?	Computer vision landmarks, spatial-temporal graph learning, Transformer sequence modeling and NLP.
Why not simple gesture recognition?	Continuous signing depends on movement and relationships among multiple joints.
Why ST-GCN?	It models graph structure and movement over time.
Why Transformer?	It models sequence context and supports text generation.
Can it translate all ISL?	The academic system should be described as supporting a defined dataset/vocabulary; broader coverage is future work.
How is privacy handled?	The architecture favors local/edge processing and avoids mandatory raw-video cloud storage.
Latency target?	Below 500 ms end-to-end as a design target, subject to measurement.

16. Presentation Truthfulness Rule

Say 'implemented' only for components that actually run. Say 'proposed' or 'planned' for designed but unfinished architecture. Say 'future scope' for capabilities not yet built and evaluated.